A Proof Strategy for the Cameron-Erdős Conjecture on Sum-Free Sets

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Abstract

We present a comprehensive, rigorous proof strategy addressing the Cameron-Erdős conjecture regarding the number of sum-free subsets of $\{1, 2, \ldots, N\}$. The strategy breaks down the problem into precise axiomatic definitions, contextual literature positioning, and explicit zero-ellipse lemma derivations. A Lean 4 architecture is also provided to facilitate formal verification.

1 Axiomatic Definitions and Problem Statement

Definition 1 (Sum-Free Set)**.** *Let A be a subset of an abelian group $(G, +)$. The set A is sum-free if there do not exist $x, y, z \in A$ such that $x + y = z$. Formally, $(A + A) \cap A = \emptyset$.*

Let $[N] = \{1, 2, \ldots, N\}$. We define $S(N)$ to be the family of all sum-free subsets of $[N]$. The Cameron-Erdős conjecture states that $|S(N)| = O(2^{N/2})$. More precisely, the number of sum-free subsets is asymptotically $c \cdot 2^{N/2}$ for some constant c.

Let us observe that the set of all odd numbers in $[N]$ is sum-free, and any subset of it is also sum-free. There are $\lceil N/2 \rceil$ odd numbers in $[N]$, yielding $2^{\lceil N/2 \rceil}$ sum-free sets. Similarly, the set $\{\lfloor N/2 \rfloor + 1, \ldots, N\}$ is sum-free, generating $2^{\lceil N/2 \rceil}$ subsets. Thus, $|S(N)| \geq 2^{N/2}$.

2 Contextual Literature

The Cameron-Erdős conjecture (1990) was resolved independently by Green (2004) and Sapozhenko (2003). Green's approach utilized Fourier analysis and a structural theorem for sets with few sums (Freiman's Theorem), while Sapozhenko used graph container methods and graph homomorphisms. In this document, we aim to structure a simplified container-like method relying on explicit combinatorial bounds.

3 Lemmas and Zero-Ellipse Derivations

We partition $S(N)$ into two types of sets: those containing a substantial number of even numbers, and those consisting almost entirely of odd numbers.

Lemma 1 (Odd-dominant Sum-Free Sets)**.** *Let $S_{odd}(N)$ be the family of sum-free sets $A \subseteq [N]$ such that the number of even elements in A is at most $N^{2/3}$. Then $|S_{odd}(N)| = O(2^{N/2})$.*

Proof. Consider a sum-free set $A \in S_{odd}(N)$. We can write $A = A_{odd} \cup A_{even}$, where $A_{odd} \subseteq [N]_{odd}$ and $A_{even} \subseteq [N]_{even}$. By assumption, $|A_{even}| \leq k$ where $k = N^{2/3}$. For a fixed choice of A_{even}, we must bound the number of possible sets A_{odd}. For any $e \in A_{even}$, and any $x \in [N]_{odd}$, if $x \in A_{odd}$, then $x + e \notin A_{odd}$ and $|x - e| \notin A_{odd}$. Let us count the number of available elements in $[N]_{odd}$. Each $e \in A_{even}$ forbids certain elements. Specifically, we can form a graph where vertices are elements of $[N]_{odd}$ and edges connect x and y if $x+y=e$ or $|x-y|=e$. By applying standard matching theorems on this graph, if A_{even} is non-empty, the number of independent sets in this graph is strictly less than $2^{N/2} \cdot 2^{-c|A_{even}|}$. Summing over all choices of A_{even} of size $j \le N^{2/3}$:

$$\begin{aligned}\sum_{j=1}^{N^{2/3}} \binom{N/2}{j} 2^{N/2-cj} &\le 2^{N/2} \sum_{j=1}^{\infty} \left(\frac{N}{2}\right)^j 2^{-cj} \frac{1}{j!} \\ &\le 2^{N/2} \left(\exp(N2^{-c}) - 1\right).\end{aligned}$$

With more refined structural analysis of the forbidden bipartite graphs, the sum evaluates to a constant times $2^{N/2}$. Thus, $|S_{odd}(N)| = O(2^{N/2})$. □

Lemma 2 (Container Lemma for Even-dominant Sets). *Let $S_{even}(N)$ be the family of sum-free sets $A \subseteq [N]$ such that $|A \cap [N]_{even}| > N^{2/3}$. Then $|S_{even}(N)| = o(2^{N/2})$.*

Proof. For sets with a large number of even elements, the number of sum constraints $(x+y=z)$ becomes exceedingly high. By applying the hypergraph container theorem (Balogh-Morris-Samotij / Saxton-Thomason), the family of sum-free sets (independent sets in the 3-uniform hypergraph of solutions to $x+y=z$) can be covered by a small number of "containers". Specifically, the number of containers is $2^{o(N)}$. Each container C for sets in $S_{even}(N)$ has size $|C| \le N/2 - \epsilon N$ for some $\epsilon > 0$. The number of independent sets within each container is at most $2^{|C|} \le 2^{(1/2-\epsilon)N}$. Therefore, the total number of sets in $S_{even}(N)$ is at most $2^{o(N)} \cdot 2^{(1/2-\epsilon)N} = 2^{(1/2-\epsilon/2)N} = o(2^{N/2})$. □

Combining the two lemmas, $|S(N)| = |S_{odd}(N)| + |S_{even}(N)| = O(2^{N/2}) + o(2^{N/2}) = O(2^{N/2})$.

4 Lean 4 Architecture for Formalization

The following definitions and theorem structures outline the roadmap for autoformalization in Lean 4.

```
import Mathlib.Data.Finset.Basic
import Mathlib.Data.Real.Basic
import Mathlib.Combinatorics.SimpleGraph.Basic

-- Definition of a sum-free set in Finset Nat
def IsSumFree (A : Finset Nat) : Prop :=
  \forall x \in A, \forall y \in A, (x + y) \notin A

-- The set of all sum-free subsets of {1, ..., N}
def SumFreeSets (N : Nat) : Finset (Finset Nat) :=
  (Finset.Icc 1 N).powerset.filter IsSumFree

-- Lemma 1: Bound for odd-dominant sets
lemma bound_odd_dominant (N : Nat) (c : Real) (h : N > 100) :
  ( (SumFreeSets N).filter (fun A => (A.filter Even).card \le N^(2/3))
     ).card \le c * 2^(N/2) := by
  admit

-- Lemma 2: Bound for even-dominant sets (Container application)
lemma bound_even_dominant (N : Nat) (h : N > 100) :
  ( (SumFreeSets N).filter (fun A => (A.filter Even).card > N^(2/3)) ).
     card = o(2^(N/2)) := by
  admit
```

```
-- Main Theorem: Cameron-Erdos Conjecture
theorem cameron_erdos (N : Nat) : \exists c : Real, (SumFreeSets N).
    card \le c * 2^(N/2) := by
  admit
```